

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and research data scientist specializing in statistical measurement, experiment design, hypothesis testing, and scalable modeling for large, complex datasets. Translates product and business questions into tractable metrics, mathematical models, and validation plans; builds reproducible Python workflows for data preparation, quality assessment, and model evaluation.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Data Workflows: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), Linux, Git, AWS EC2, HPC/SLURM; data restructuring, quality validation, and reproducible pipelines; SQL/MySQL (working knowledge)

Statistical Measurement and Modeling: Experimental design, causal inference (fixed effects, propensity-score matching), hypothesis and sequential testing, regression (GLM/GLS), predictive modeling, Monte Carlo simulation, model selection and validation, performance evaluation, missing-data analysis

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Translated a business monitoring question into explicit hypotheses and measurable decision rules, developing and validating a Sequential Probability Ratio Test for binary risk indicators.
- Modeled the cumulative log-likelihood-ratio process as a random walk and applied Brownian-motion approximation and dynamic-programming probability propagation to evaluate boundary-crossing probabilities, stopping-time distributions, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM regression benchmarking, model validation, and performance evaluation on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations, translating technical results into clear actions for business stakeholders.

RESEARCH AND ANALYTICAL PROJECTS

Ph.D. Dissertation Research - Statistical Measurement, Data Quality, and Scalable Modeling

Sep 2024 - Present

- Measurement Methodology:** Designed scale- and frequency-resolved tests to identify model misspecification and quantify how missingness, noise, and dependence affect statistical estimates.
- Data Preparation and Quality:** Built workflows for observation masking, quality filtering, nearest-neighbor construction, dependency-aware ordering, and irregular-grid resampling for noisy, partially observed satellite data.
- Scalable Model Development:** Profiled regression mean structures with GLS and calibrated flexible Gaussian-process models using reusable Python/PyTorch pipelines, L-BFGS optimization, CPU/GPU execution, and restartable HPC jobs for up to 145,008 observations per day.

Economic Policy Research - EITC and Household Labor Supply

Hanyang University

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household covariates.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Linear Algebra, Statistical Computing, Stochastic Processes, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

Languages: Korean (Native); English (Fluent); Chinese (Basic)